

Cloud Computing Midterm Preparation

40 Scenario-Based Questions
with Model Answers

Topics Covered

- Introduction to Cloud Computing
 - Virtualization
 - Docker
 - Storage and Message Brokers
-

Exam Focus:

Analytical, scenario-based and comparison-type questions.

For Docker, special emphasis is placed on:

- underlying mechanics
- practical importance in cloud architecture
- essential commands

Cloud Computing & Virtualization

Midterm Preparation

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1. Introduction to Cloud Computing

Question 1

An online shopping company normally receives 5,000 requests per hour. During a festival sale, traffic suddenly increases to 100,000 requests per hour. After the sale, traffic returns to normal.

The company does not want to purchase 20 additional physical servers that will remain mostly idle.

- (a) Which cloud characteristic solves this problem?
- (b) Explain how it solves the problem.
- (c) Which deployment approach could be useful if the company keeps sensitive systems privately but uses public resources during the sale?

Answer:

(a) Rapid Elasticity.

(b) The company can rapidly provision additional resources when demand increases and release them when demand decreases. This prevents the company from purchasing permanent hardware for temporary demand.

(c) Hybrid Cloud.

A Hybrid Cloud allows sensitive workloads to remain in a private environment while public-cloud resources can be used for temporary peak demand.

Question 2

Five hospitals want to share cloud infrastructure. They have similar requirements for:

- patient-data security
- healthcare compliance
- access control
- data policies

They do not want to use a completely open public cloud, but building five separate private clouds would be expensive.

Which deployment model would you recommend? Why?

Answer:

Community Cloud.

The hospitals have common security, compliance and policy requirements. They can therefore share infrastructure designed specifically for their community.

This avoids each hospital independently building and maintaining a completely separate private cloud.

Question 3

A startup has three developers. They want to deploy a web application but do not want to:

- manage Linux
- install operating-system patches
- configure servers
- maintain physical infrastructure

They only want to upload their application code.

Which service model is most appropriate? Explain.

Answer:

PaaS – Platform as a Service.

The cloud provider manages the underlying infrastructure and platform/operating-system environment, while the developers mainly focus on application code.

Google App Engine is an example of this type of service.

Question 4

A company rents computing resources from a cloud provider.

The company chooses:

- the operating system
- installed software
- security configuration
- applications

The provider mainly supplies virtualized compute, storage and networking infrastructure.

Which service model is this? Why?

Answer:

IaaS – Infrastructure as a Service.

The customer receives infrastructure resources and retains substantial control over the operating system and software.

The cloud provider manages the underlying physical infrastructure.

Question 5

A university purchases an online email service.

Students simply log in through a browser or mobile application.

The university does not manage:

- servers
- operating systems
- email software installation
- storage infrastructure

(a) Which service model is this?

(b) Which deployment model could commonly provide it?

Answer:

(a) SaaS – Software as a Service.

The complete application is provided to the users.

(b) Public Cloud is a common deployment model because the service can be accessed by users through the Internet.

Question 6

A company buys one powerful physical server and places it in its office.

The server:

- belongs entirely to the company
- cannot automatically scale
- is not shared
- requires manual hardware upgrades
- has no usage-based provisioning

The IT manager claims:

“It is cloud computing because the server is powerful.”

Do you agree? Justify your answer using the characteristics of cloud computing.

Answer:

No.

A powerful physical server by itself does not constitute cloud computing.

The scenario does not demonstrate important cloud characteristics such as:

- on-demand self-service
- resource pooling
- rapid elasticity
- measured service

Therefore, simply having a powerful server is not sufficient.

Question 7

A cloud provider runs thousands of customers' virtual machines on a large pool of physical servers.

Customers do not normally know the exact physical server on which their VM is running.

- (a) Which NIST characteristic is demonstrated?
- (b) Why is this beneficial to the provider?

Answer:

Resource Pooling.

The provider pools physical resources and dynamically assigns and reassigns them to customers according to demand.

This improves resource utilization because physical resources do not need to be permanently dedicated to one customer.

Question 8

Two companies use the same cloud provider.

Company A uses:

- 2 TB storage
- 500 GB network transfer
- 10 virtual CPUs

Company B uses:

- 500 GB storage
- 100 GB network transfer
- 2 virtual CPUs

The provider charges them different amounts based on their consumption.

Which cloud characteristic is demonstrated? Explain.

Answer:

Measured Service.

The provider meters resource consumption such as processing, storage and bandwidth. Customers can therefore be monitored and charged according to their resource usage. A simple analogy is a taxi meter: greater usage results in greater charges.

Question 9

A bank wants:

- customer databases to remain under strict organizational control
- public-cloud resources for less-sensitive web traffic
- additional capacity during periods of high demand

Which deployment model would you recommend? Justify your answer.

Answer:

Hybrid Cloud.

Sensitive workloads can remain in a private environment, while less-sensitive or highly elastic workloads can use public-cloud resources. This provides a balance between control/security and scalability.

Question 10

A company needs to decide:

1. whether to use Public, Private or Hybrid Cloud
2. whether developers should manage the operating system

A student says:

“Public Cloud and IaaS are alternatives because both describe the cloud.”

Is the student correct? Explain.

Answer:

No.

They describe two different dimensions.

Deployment model describes how/where the cloud environment is organized:

Public, Private, Community, Hybrid

Service model describes the responsibility boundary between the provider and customer:

IaaS, PaaS, SaaS

Therefore, combinations such as **Public + IaaS** or **Hybrid + IaaS** are possible.

2. Virtualization

Question 11

A large cloud data center needs to run thousands of virtual machines.

Two hypervisors are available:

- Hypervisor A runs directly on physical hardware.
- Hypervisor B runs as an application on Windows/Linux.

Which should the data center prefer and why?

Answer:

Hypervisor A – Type 1 / Bare-metal Hypervisor.

It runs directly on physical hardware and does not require an additional host operating-system layer. This generally reduces overhead and makes Type 1 hypervisors well suited for large data centers.

Question 12

A student wants to run Ubuntu inside Windows on a personal laptop without replacing the existing Windows installation.

Which hypervisor type would be more appropriate? Why?

Answer:

Type 2 / Hosted Hypervisor.

A Type 2 hypervisor runs on top of an existing host operating system.

It is convenient for personal computers, development and testing.

Examples include VirtualBox and VMware Workstation.

Question 13

A physical server has:

- CPU
- RAM
- SSD

A hypervisor is installed on it. Three VMs are created:

- VM1 – Windows
- VM2 – Ubuntu
- VM3 – Linux

Describe the three-layer virtualization stack and explain what happens at each layer.

Answer:

The stack is:

```

+-----+
| VM1: Windows      |
| VM2: Ubuntu       |
| VM3: Linux        |
+-----+
|      Hypervisor   |
+-----+
| Physical Hardware  |
| CPU | RAM | SSD    |
+-----+

```

Physical hardware provides the actual resources.

Hypervisor abstracts and manages those physical resources.

Guest VMs receive virtualized hardware and run their own guest operating systems.

Question 14

A company currently has ten physical servers.

Each server uses only around 15% CPU and runs one application.

The company wants to reduce:

- hardware costs
- electricity consumption
- cooling requirements

How could server virtualization solve this problem?

Answer:

Multiple workloads can be consolidated into separate VMs running on fewer physical servers.

For example:

Physical Server

```

|
+-- VM 1 -> Application A
+-- VM 2 -> Application B
+-- VM 3 -> Application C
+-- VM 4 -> Application D

```

This increases hardware utilization and reduces the number of physical machines, electricity consumption and cooling requirements.

Question 15

A cloud administrator creates:

- virtual networks
- virtual switches
- subnets
- routing rules
- firewall policies

These are configured through software instead of manually changing physical network devices.

What concept is being demonstrated? Explain.

Answer:

Network Virtualization / Software-Defined Networking (SDN).

Network functionality is abstracted from the physical hardware and can be controlled through software.

This makes network configuration more flexible, programmable and easier to modify.

Question 16

A company has 50 physical servers.

Every time administrators need a new subnet, they must manually reconfigure several physical networking devices.

The company wants to create networks dynamically through software.

Would network virtualization help? How?

Answer:

Yes.

Network virtualization abstracts network resources and allows virtual networks, routing and policies to be created or modified through software.

This reduces dependence on manually changing physical network infrastructure.

Question 17

An organization has several physical storage systems but presents them to administrators as one logical storage pool.

Which type of virtualization is being used?

Why is it useful?

Answer:

Storage Virtualization.

Multiple physical storage resources are abstracted into a logical storage pool.

This simplifies storage management and allows capacity to be allocated more flexibly.

Question 18

A company wants employees to access centrally hosted desktop environments from:

- laptops
- tablets
- thin clients

The actual desktop environment runs centrally.

Which virtualization technology is being used?

Answer:

Desktop Virtualization.

The desktop environment is hosted centrally while users access it through endpoint devices.

A thin client can act primarily as the access device while the desktop workload is processed centrally.

Question 19

A company wants employees to use a particular business application without performing a traditional full installation on every employee's computer.

Which virtualization type is most appropriate?

How does it differ from desktop virtualization?

Answer:

Application Virtualization.

It focuses on delivering or running an application separately from the traditional local installation model.

Desktop virtualization, on the other hand, provides an entire desktop environment rather than only an individual application.

Question 20

A student says:

“Virtualization and cloud computing are the same thing because AWS uses virtual machines.”

Evaluate this statement.

Answer:

The statement is incorrect.

Virtualization is a technology that abstracts physical resources such as CPU, memory, storage and networking.

Cloud computing is a model for delivering computing resources and services.

Virtualization can be used to build cloud infrastructure, but virtualization alone does not guarantee cloud characteristics such as:

- on-demand self-service
- broad network access
- resource pooling
- rapid elasticity
- measured service

A useful way to remember this is:

Virtualization = Technology
Cloud Computing = Service/Delivery Model

3. Docker

Question 21

A company wants to deploy 50 small microservices.

Option A creates 50 VMs, each containing a complete guest OS.

Option B creates 50 Docker containers sharing the host kernel.

Which option is likely to use fewer resources? Why?

Answer:**Option B – Docker Containers.**

Containers share the host operating-system kernel instead of requiring a complete guest OS for every application.

Therefore, containers generally require less memory and storage and start faster than VMs.

Conceptually:

VM:

Hardware

|

Hypervisor

|

Guest OS

|

Application

Container:

Hardware

|

Host OS Kernel

|

Container

|

Application

Question 22

A developer executes:

```
docker run nginx
```

The container starts much faster than a traditional VM.

Explain why using the underlying architecture rather than simply saying “Docker is faster.”

Answer:

A VM normally needs to boot a complete guest operating system.

A Docker container starts an isolated application/process while sharing the host operating-system kernel.

Therefore:

VM:

Hardware -> Hypervisor -> Guest OS -> Application

Container:

Hardware -> Host OS Kernel -> Container -> Application

Because the container does not need to boot a complete guest OS, startup is generally much faster.

Question 23

A company runs:

```
docker run -d --name web1 nginx
docker run -d --name web2 nginx
```

Are these the same container?

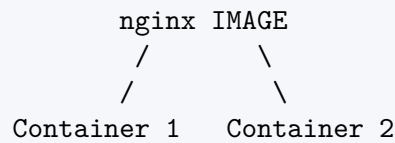
Explain the relationship between an image and a container.

Answer:

No.

They are two separate containers created from the same nginx image.

The relationship is:



The **image** is the reusable template, while each **container** is an individual runtime instance.

Question 24

A student executes:

```
docker run nginx
```

Describe the sequence involving:

- Docker CLI
- Docker daemon
- image
- container

Answer:

1. The user enters the command through the **Docker CLI**.
2. The CLI communicates with the **Docker daemon**.
3. The daemon checks whether the nginx image exists locally.
4. If necessary, Docker pulls the image from a registry.
5. Docker creates a container from the image.
6. The daemon starts the container.

Therefore:

CLI → Docker Daemon → Image → Container

Question 25

A student successfully executes:

```
docker run -d -p 8080:80 nginx
```

They then try:

```
docker run -d -p 8080:80 nginx
```

Docker reports that the port is already allocated.

Why did this happen?

How would you solve it?

Answer:

Both containers are trying to use the same **host port 8080**.

Use a different host port for the second container:

```
docker run -d -p 8081:80 nginx
```

Now:

Host 8080 -> Container 1 port 80

Host 8081 -> Container 2 port 80

The container port can remain 80; the host ports must be different.

Question 26

A Redis container stores important application data.

The developer removes the container and discovers that the data disappears when a new container is created.

What architectural mistake was made?

What should have been used instead?

Answer:

The data was stored inside the container's writable layer.

Container storage is tied to the container lifecycle.

For persistent data, a **Docker Volume** should be used.

Conceptually:

Container

|

v

Docker Volume

|

v

Persistent Data

A volume allows data to survive even if the container itself is removed and recreated.

Question 27

A Flask container needs to communicate with a Redis container using the name:

`redis-db`

DNS resolution fails because the containers were not placed on the same suitable user-defined bridge network.

How should the architecture be changed?

Answer:

Both containers should be connected to the same **user-defined bridge network**.

For example:

```
docker network create app-net

docker run -d --name redis-db \
    --network app-net redis

docker run -d --name web \
    --network app-net flask-app
```

Docker's user-defined networking provides container-to-container connectivity and DNS-based service discovery.

The application can therefore resolve the Redis service using its container/service name rather than relying on a changing IP address.

Question 28

A developer creates this Dockerfile:

```
FROM python:3.11-slim
COPY . .
RUN pip install -r requirements.txt
```

Every time the developer changes one line of application code, Docker installs all dependencies again.

How could the Dockerfile be reorganized to make better use of Docker's layer cache?

Answer:

Copy the dependency file first, install dependencies, and then copy the application source code.

For example:

```
FROM python:3.11-slim

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir \
    -r requirements.txt

COPY app.py .
```

Now, if only `app.py` changes, Docker can reuse the previous dependency-installation layer. This makes builds faster.

Question 29

A web application consists of:

- Flask
- Redis
- Adminer

The developer wants to define:

- ports

- volumes
- environment variables
- dependencies

and start everything using one command.

What Docker feature should be used? Explain.

Answer:

Docker Compose.

The services can be defined together in a `compose.yaml` file.

The file can specify:

- images/build instructions
- ports
- volumes
- environment variables
- dependencies
- networks

The complete stack can then be started with:

```
docker compose up -d --build
```

This is particularly useful for multi-container applications.

Question 30

A student's machine is almost full of unused:

- containers
- images
- volumes

The student executes:

```
docker system df
```

and considers:

```
docker system prune -a --volumes -f
```

What do these commands do?

What is the main danger?

Answer:

`docker system df` reports Docker's disk-space usage.

The command:

```
docker system prune -a --volumes -f
```

removes unused Docker resources, including unused images, containers and volumes.

The main danger is deleting an unused-but-still-needed volume containing persistent data.

Therefore, pruning with `-volumes` should be performed carefully.

4. Storage and Message Brokers

Question 31

A video-sharing company stores:

- millions of videos
- images
- backups
- logs

The data is mostly unstructured and must scale massively. The company wants to access the data through an API.

Which storage type should be selected? Why?

Answer:

Object Storage.

Object storage is designed for large-scale unstructured data.

It is appropriate for:

- videos
- images
- backups
- logs

Objects are generally accessed through APIs and stored in buckets or similar logical containers.

Question 32

A university wants a shared departmental drive with:

Teachers/

Students/

Exams/

Research/

Many computers need simultaneous access through familiar file/folder semantics.

Which storage type is appropriate?

Answer:

File Storage.

File storage provides hierarchical file and folder organization.

It is appropriate for:

- shared drives
- home directories
- collaborative file access

Users can interact with the storage using familiar file-system semantics.

Question 33

A banking application performs thousands of transactions per second.

It requires:

- low latency
- frequent reads and writes

- disk-like access
- database workloads

Which storage type should be selected?

Answer:

Block Storage.

Block storage provides disk-like volumes and is appropriate for:

- databases
- transactional applications
- low-latency workloads
- frequent read/write operations

Question 34

A small business has one server.

It wants a simple backup disk directly connected to that server. Other machines do not need to access the storage.

Should it use:

- DAS
- NAS
- SAN

Why?

Answer:

DAS – Direct-Attached Storage.

The storage is directly connected to one host.

DAS is simple and relatively inexpensive and is appropriate when storage is primarily required by a single server.

Question 35

A company wants 100 employees to access shared documents through the normal LAN using familiar file-sharing semantics such as a network drive.

Should it use DAS, NAS or SAN?

Explain.

Answer:

NAS – Network-Attached Storage.

NAS provides shared file storage over a network.

It is appropriate when many users or devices need to access common files.

Typical protocols include:

- SMB
- NFS

Question 36

An enterprise has multiple database servers requiring high-performance block storage.

The company is willing to pay more for:

- high performance
- centralized storage
- multiple-server access
- a dedicated storage network

Which architecture should be considered?

Answer:

SAN – Storage Area Network.

SAN provides block-level storage over a dedicated high-speed storage network.

It is appropriate for demanding enterprise workloads such as databases and systems requiring high-performance centralized storage.

Question 37

A payroll service creates one payment transaction.

The transaction must be processed by exactly one worker.

There are currently five worker instances.

If one worker is offline, the transaction should wait until another worker can process it.

Which messaging pattern should be used? Why?

Answer:

Point-to-Point / Queue.

The producer places the message into a queue.

One worker consumes each message.

Conceptually:

Producer

|

v

Queue

/ | \

W1 W2 W3

Only one worker processes a particular message.

If a worker is temporarily unavailable, another worker can eventually consume the queued message.

This is appropriate for payroll and financial transactions.

Question 38

An online store generates the event:

OrderPlaced

Three independent services need to receive the event:

- Email Service
- SMS Service
- Analytics Service

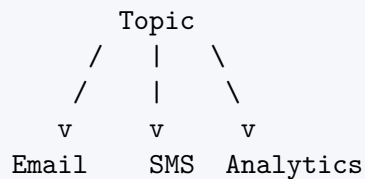
The producer should not need to know which services exist.

Which messaging pattern should be used? Explain.

Answer:**Publish/Subscribe.**

The order service publishes the event to a topic.

Multiple independent subscribers receive the event:



This provides one-to-many delivery and loose coupling.

New subscribers can be added without modifying the producer.

Question 39

A web server synchronously calls an image-processing service through REST.

When the image-processing service goes offline, users receive errors because the request cannot be processed.

How could a message broker improve this architecture?

Answer:

Instead of:

Web Server ----REST----> Worker

use:

Web Server ----> Queue/Broker ----> Worker

The web server places a message in the broker and can continue without waiting for the worker to finish.

If the worker is temporarily unavailable, the message can remain in the queue until the worker becomes available.

Benefits include:

- asynchronous communication
- decoupling
- improved reliability
- better handling of temporary failures

Question 40

An application allows users to upload photos.

After a user uploads a photo, the system must:

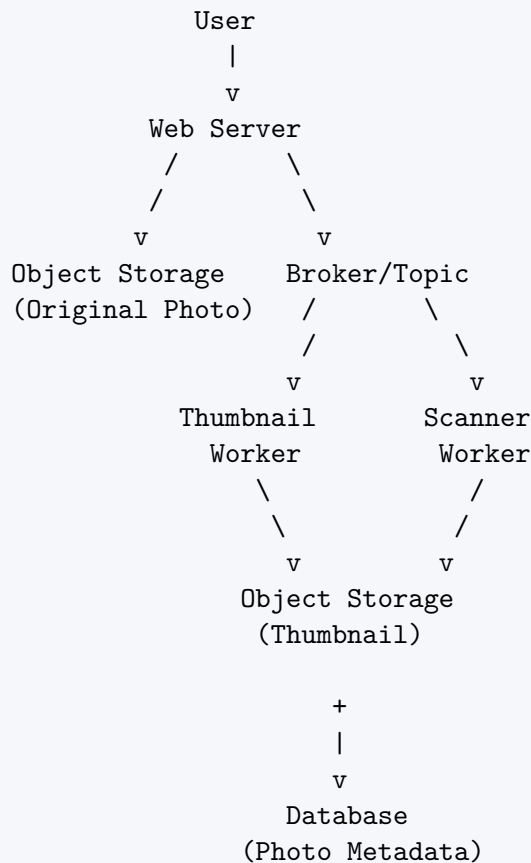
1. store the original photo
2. create a thumbnail
3. scan the image for malicious content
4. save information about the photo
5. support multiple processing workers
6. continue working even if a worker is temporarily offline

Design the storage and messaging architecture.

Explain which storage type and messaging pattern you would use and justify your choices.

Answer:

A suitable architecture is:

**Original Photo: Object Storage**

The original photo is unstructured media data, making object storage appropriate.

Thumbnail: Object Storage

The thumbnail is also a media object and can be stored in object storage.

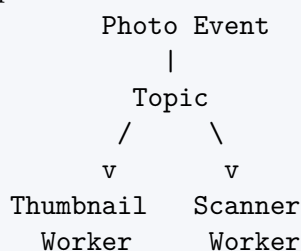
Photo Metadata: Database

Information such as filename, user ID, upload time and object location is structured data and should be stored in a database.

Messaging: Publish/Subscribe

The web server can publish a “new photo” event to a topic.

Multiple independent services can subscribe:



This is preferable to making the web server synchronously call every processing service.

The broker provides:

- asynchronous processing
- loose coupling
- scalability
- better failure handling

If a worker temporarily goes offline, the message can remain available for processing rather than causing the entire upload request to fail.

Last-Minute Revision Strategy

For scenario-based questions, do not memorize only definitions.

Use this process:

1. Identify the requirement.

Ask what the scenario is actually trying to achieve:

- scalability?
- control?
- resource efficiency?
- persistence?
- high performance?
- shared access?
- asynchronous processing?
- fault tolerance?

2. Map the requirement to the concept.

For example:

High traffic spike	Rapid Elasticity
Sensitive + Public workloads	Hybrid Cloud
Large data center	Type 1 Hypervisor
Lightweight deployment	Containers
Persistent container data	Docker Volume
Images/videos/backups	Object Storage
Shared folders	File Storage
Database disk	Block Storage
One worker per message	Queue
One event to many services	Pub/Sub

3. Justify the choice.

Do not simply write:

“The answer is Object Storage.”

Instead write:

“Object Storage is appropriate because the application stores large amounts of unstructured media such as images and videos and needs scalable API-based access.”

4. For comparisons, mention the architectural difference.

For example:

- Type 1 vs Type 2
- VM vs Container
- Object vs File vs Block
- DAS vs NAS vs SAN
- Queue vs Pub/Sub
- IaaS vs PaaS vs SaaS

Highest-Priority Questions

If time is limited, practice these first:

1. Question 1 – Rapid Elasticity + Hybrid Cloud
2. Question 2 – Community Cloud
3. Question 10 – Deployment Model vs Service Model
4. Question 11 – Type 1 vs Type 2
5. Question 13 – Three-Layer VM Architecture
6. Question 20 – Virtualization vs Cloud Computing
7. Question 22 – Docker Underlying Mechanics
8. Question 26 – Docker Volumes and Persistence
9. Question 28 – Dockerfile Layers and Caching
10. Question 40 – Integrated Storage + Message Broker Architecture

Quick Memory Sheet

Cloud Computing

Characteristic	Meaning
On-Demand Self-Service	Provision resources when needed
Broad Network Access	Accessible over networks
Resource Pooling	Shared provider resources
Rapid Elasticity	Scale up/down quickly
Measured Service	Usage is monitored/metered

Deployment Models

- **Public** – provider-owned infrastructure shared among customers
- **Private** – dedicated to one organization
- **Community** – shared by organizations with common requirements
- **Hybrid** – combination of environments

Service Models

Model	Customer Mainly Manages	Example
IaaS	OS + applications	Amazon EC2
PaaS	Application/code	Google App Engine
SaaS	Mainly usage/configuration	Webmail

Virtualization

Type 1:
 Hardware
 |
 Hypervisor
 |
 VMs

Type 2:
 Hardware
 |
 Host OS
 |
 Hypervisor
 |
 VMs

Docker

Docker CLI
 |
 v
 Docker Daemon
 |
 +----- Image
 |

```

+---- Container
|
+---- Volume
|
+---- Network

```

Important commands:

```

docker images
docker ps
docker ps -a
docker run
docker start
docker stop
docker rm
docker pull
docker exec
docker logs
docker build
docker network create
docker volume create
docker compose up -d
docker compose down
docker system df
docker system prune -a --volumes -f

```

Storage

Storage	Typical Use
Object	Images, videos, backups, logs
File	Shared folders, home directories
Block	Databases, low-latency disk workloads

DAS / NAS / SAN

DAS	Directly attached	Single host
NAS	Network file storage	Shared files
SAN	Dedicated storage network	Enterprise block storage

Message Brokers

Queue / Point-to-Point	One message → one worker
Publish/Subscribe	One event → many subscribers